# Magneto-statics

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| | |
|---|---|
| $\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\mathbf{a}_r$ | $\mathbf{H} = \oint \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$ |
| ρl dl = ρs ds = ρv dv | IdL = KdS = JdV |
| $\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$ | $\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi$ |

- Find H due to infinitely long current carrying conductor
- Due to symmetry – No variation with z

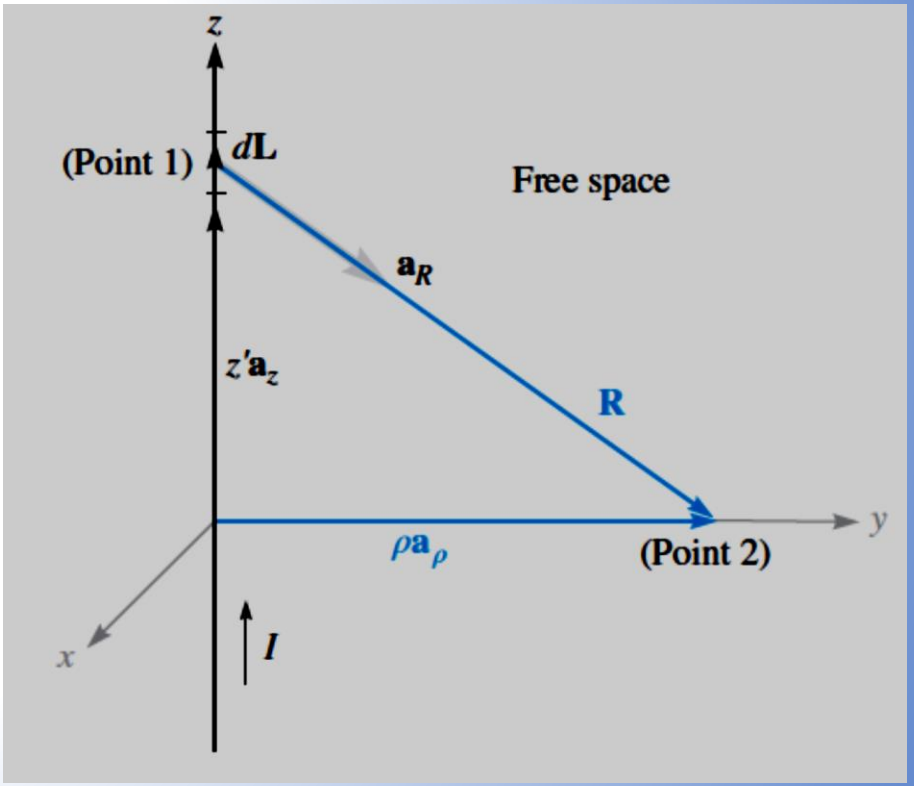
  - No variation with $\phi$

$$\mathbf{R}_{12} = \mathbf{r} - \mathbf{r}' = \rho \mathbf{a}_\rho - z' \mathbf{a}_z$$

$$\mathbf{a}_{R12} = \frac{\rho \mathbf{a}_\rho - z' \mathbf{a}_z}{\sqrt{\rho^2 + z'^2}}$$

$$d\mathbf{L} = dz' \mathbf{a}_z$$

$$d\mathbf{H} = \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{I d\mathbf{L} \times \mathbf{R}}{4\pi R^3}$$



$$d\mathbf{H} = \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{Id\mathbf{L} \times \mathbf{R}}{4\pi R^3}$$

$$d\mathbf{H}_2 = \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{H}_2 = \int_{-\infty}^{\infty} \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

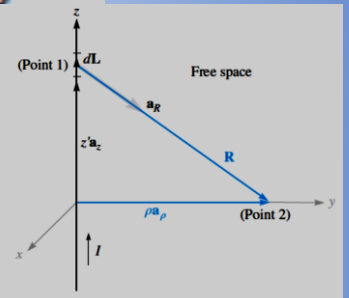
$$= \frac{I}{4\pi}\int_{-\infty}^{\infty} \frac{\rho dz'\mathbf{a}_\phi}{(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{H}_2 = \frac{I\rho\mathbf{a}_\phi}{4\pi}\int_{-\infty}^{\infty} \frac{dz'}{(\rho^2 + z'^2)^{3/2}}$$ [Refer next page}

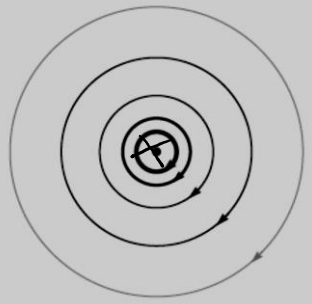
$$= \frac{I\rho\mathbf{a}_\phi}{4\pi} \frac{z'}{\rho^2\sqrt{\rho^2 + z'^2}}\Bigg|_{-\infty}^{\infty}$$

Split the limits

$$\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi$$




H Streamlines are into the page →



$a_\rho \rightarrow a_\phi \rightarrow a_z$

Note: For electrostatics

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$

$$I = \int \frac{1}{(x^2 + a^2)^{3/2}}\, dx$$

Let's simplify it by making a substitution...

$$x = a \tan t$$

Taking derivative of both sides...

$$dx = a \sec^2 t \; dt$$

$$I = \int \frac{a \sec^2 t}{(a^2 \tan^2 t + a^2)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{(a^2 (\tan^2 t + 1))^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 (\tan^2 t + 1)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 (\sec^2 t)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 \sec^3 t}\, dt$$

$$I = \int \frac{1}{a^2 \sec t}\, dt$$

$$I = \int \frac{\cos t}{a^2}\, dt$$

$$I = \frac{1}{a^2} \int \cos t \, dt$$

$$I = \frac{\sin t}{a^2} + C$$

We had initially substituted $x = a \tan t$

That means $\tan t = \frac{x}{a}$

So, we have $\sin t = \frac{x}{\sqrt{a^2 + x^2}}$

$$I = \frac{\frac{x}{\sqrt{a^2 + x^2}}}{a^2} + C$$

$$I = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

$$I = \int \frac{1}{(x^2 + a^2)^{3/2}}\, dx \quad = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

[You can take it as a standard result/formula]

In free space, we define the *magnetic flux density* **B** as

$$\mathbf{B} = \mu_0 \mathbf{H} \quad \text{(free space only)}$$

webers per square meter ($Wb/m^2$) or tesla (T).

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

The name given to $\mu_0$ is the *permeability* of free space.

$$\mu = \mu_0 \mu_r$$

**Problem 7.1 : Find the incremental field $\Delta \bar{H}_2$ at $P_2$ caused by a source at $P_1$ of $I_1 \Delta \bar{L}_1 =$ : (a) $2\pi \bar{a}_z \mu$ A · m, given $P_1(4, 0, 0)$ and $P_2(0, 3, 0)$; (b) $2\pi \bar{a}_z \mu$ A · m, given $P_1$ (4, −2, 3) and $P_2$ (0, 3, 0); (c) $2\pi (0.6\bar{a}_x - 0.8\bar{a}_y) \mu$ A · m, given $P_1$ (4, −2, 3) and $P_2$ (1, 3, 2).**

(a) $$d\bar{H}_2 = \frac{I\,\overline{dL} \times \bar{a}_{12}}{4\pi d^2}$$

Point 2 is that point at which magnetic field intensity is desired = (0, 3, 0)

Point1 is that point at which differential current element is located= (4, 0, 0)

$$\bar{R}_{12} = (0 - 4)\bar{a}_x + (3 - 0)\bar{a}_y = -4\bar{a}_x + 3\bar{a}_y \quad (P_2 - P_1)$$

$$d = |\bar{R}_{12}| = \sqrt{(-4)^2 + (3)^2} = 5$$

$$\therefore d\bar{H}_2 = \frac{(2\pi 10^{-6}\bar{a}_z)}{4\pi (5)^2} \times \left[\frac{-4\bar{a}_x + 3\bar{a}_y}{5}\right] = -12\bar{a}_x - 16\bar{a}_y \text{ nA/m}$$

(b) $\bar{R}_{12} = (0 - 4)\,\bar{a}_x + (3 + 2)\,\bar{a}_y + (0 - 3)\,\bar{a}_z$

$= -4\,\bar{a}_x + 5\,\bar{a}_y - .3\,\bar{a}_z$

$P_2\,(0, 3, 0)$

$P_1\ (4, -2, 3)$

$$d = |\bar{R}_{12}| = \sqrt{(-4)^2 + (5)^2 + (-3)^2} = \sqrt{50} = 7.071$$

$$\therefore d\bar{H}_2 = \frac{(2\pi\, 10^{-6}\,\bar{a}_z)}{4\pi\,(7.071)^2} \times \left[\frac{-4\,\bar{a}_x + 5\,\bar{a}_y - 3\,\bar{a}_z}{7.071}\right]$$

$= -7.071\,\bar{a}_x - 5.657\,\bar{a}_y$ n A/m

(c) $\bar{R}_{12} = (1 - 4)\,\bar{a}_x + (3 + 2)\,\bar{a}_y + (2 - 3)\,\bar{a}_z$

$= -3\,\bar{a}_x + 5\,\bar{a}_y - \bar{a}_z$

$P_2\ (1, 3, 2)$

$P_1\ (4, -2, 3)$

$$d = |\bar{R}_{12}| = \sqrt{(-3)^2 + (5)^2 + (-1)^2} = \sqrt{35} = 5.916$$

$$\therefore d\bar{H}_2 = \left[\frac{2\pi\,(0.6\,\bar{a}_x - 0.8\,\bar{a}_y)\,10^{-6}}{4\pi\,(5.916)^2}\right] \times \left[\frac{-3\,\bar{a}_x + 5\,\bar{a}_y - \bar{a}_z}{5.916}\right]$$

$= 1.932\,\bar{a}_x + 1.44\,\bar{a}_y + 1.449\,\bar{a}_z$ nA/m


Thank
you